

GLM Investigation Report:

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HDS-5230 : High Performance Computing

Module /framework /package	Name and a brief description of the algorithm	An example of a situation where using the provided GLM implementation provides superior performance compared to that of base R or its equivalent in Python
Base R (stats library)	Iteratively Reweighted Least Squares (IRLS) / Fisher Scoring: An iterative algorithm that approximates the GLM fitting problem as a sequence of weighted least squares problems. Equivalent to Fisher Scoring in most cases, using the expected information matrix.	Suitable for small to moderately sized datasets that fit into memory, where standard GLM analysis and statistical inference are the primary goals.
biglm (R)	Incremental IRLS: Adapts the IRLS algorithm to process data in chunks, allowing for fitting GLMs to datasets that are too large to fit into memory.	When dealing with massive datasets that exceed the available RAM on a single machine, the primary need is to obtain parameter estimates without requiring distributed computing .
h2o (R)	Optimization with Elastic Net Regularization: Employs an optimization algorithm (likely coordinate descent or similar) suitable for distributed execution, with built-in L1 and L2 regularization.	For very large datasets that can be distributed across a cluster , and when regularization is desired for feature selection and preventing overfitting .
glmnet (R)	Cyclical Coordinate Descent: Iteratively optimizes the objective function over each parameter with others fixed, particularly efficient for penalized maximum likelihood estimation with sparse data .	When fitting GLMs with L1, L2, or elastic net penalties, especially for high-dimensional datasets where sparsity is expected .
Dask ML (Python)	ADMM, Gradient Descent, L-BFGS, Newton's Method, Proximal Gradient: Offers a variety of optimization algorithms that can be	When working with large datasets that can be distributed across multiple cores or machines using Dask, and when specific optimization algorithms like ADMM (Alternating Direction Method of Multipliers),

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	chosen based on the specific GLM problem and dataset characteristics.	when we are dealing with objective function and solution space that is in convex form .
Spark R (spark.glm)	Distributed IRLS (using Spark MLLib's Gradient Descent or L-BFGS) : Utilizes IRLS for GLM fitting, with the underlying weighted least squares steps powered by Spark MLLib's distributed optimization algorithms.	For extremely large datasets and distributed computing environments managed by Apache Spark, where scalability and parallel processing are essential .
Scikit-learn (Python)	liblinear, newton-cg, lbfgs, sag, saga : Provides a range of optimization solvers for linear models, including those applicable to GLMs (like Logistic Regression), with options for different regularization types and dataset sizes.	For moderately sized datasets where speed, ease of use , and access to various efficient optimization algorithms and regularization techniques are desired.

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